

Snooker Intelligent Analyzer & Shot Recommendation System

By

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Snooker Intelligent Analyzer &
Shot Recommendation System

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A project report submitted to the
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in partial fulfillment of the requirement for the
Choose an item.

Faculty of Computing and Information Technology
Tunku Abdul Rahman University of Management and Technology
Kuala Lumpur

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Project Title Abstract

Abstract

Snooker is a precision-based cue sport that demands high levels of strategic thinking, spatial awareness, and accurate shot execution. While professional players rely on extensive experience to evaluate table conditions and plan optimal shots, amateur and intermediate players often

struggle due to the lack of real-time, objective feedback during gameplay. Existing snooker-related applications are largely limited to manual scorekeeping and fail to provide meaningful analytical support. Moreover, applying computer vision to snooker presents significant technical challenges, including small object detection, ball occlusion, motion blur, and color distortion under variable lighting conditions.

This project proposes the development of a Snooker Intelligent Analyzer & Shot Recommendation System, an AI-driven solution designed to deliver real-time ball detection, automated rule enforcement, and game state analysis using mobile devices. The system employs enhanced YOLO-based object detection architectures optimized for small object detection to accurately identify and classify all snooker balls on a full-size table. Specialized color space transformations and illumination-invariant techniques are applied to reliably distinguish between visually similar balls under diverse lighting environments.

Beyond visual detection, a Game State Analysis Engine (GSAE) is implemented to function as a digital referee by enforcing official World Professional Billiards and Snooker Association (WPBSA) rules. This engine automates score tracking, validates legal shots, detects fouls, and calculates remaining points on the table, significantly reducing the cognitive load on players. The system is optimized for on-device deployment using TensorFlow Lite with model quantization to achieve real-time performance on Android smartphones.

Experimental evaluations demonstrate that the proposed system achieves high detection accuracy while maintaining low latency suitable for real-time gameplay analysis. In conclusion, the project successfully bridges advanced computer vision techniques with practical snooker training needs, democratizing access to professional-grade analytical feedback and contributing to the advancement of intelligent sports analytics systems.

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Project Title Acknowledgement

Acknowledgement

This contains acknowledgement to those who have contributed directly or indirectly to the completion of the project. Usually the people to be acknowledged include the project supervisor(s), moderators, family, and those who have given assistance and supports to ensure the success of the project.

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Chapter 1

Introduction

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1 Introduction

Snooker is a cue sport that requires a high level of accuracy, strategic planning, and reliable performance. Unlike many other sports that prioritize physical fitness, snooker is often described as "chess with balls," demanding mental acuity and geometrical understanding from its players. For amateur and intermediate players, the problem often stems from the difficulty in accurately assessing the condition of the table and choosing the best shot from among many possibilities. While professional players rely on years of experience to visualize angles and strategies, beginners often lack immediate, objective feedback during their practice.

Recent advancements in Artificial Intelligence and Computer Vision have revolutionized sports analytics, providing tools for player tracking and trajectory analysis. However, the visual environment of a snooker table presents a "hostile" scenario for standard object detection

algorithms. Variable lighting conditions frequently cause color distortion, making it difficult to distinguish between Red, Pink, and Brown balls, while ball occlusion in clusters poses significant challenges for standard detectors. Additionally, standard architectures like YOLO often struggle with "Small Object Detection" (SOD) because snooker balls occupy a minute fraction of the visual field, leading to feature dilution and failures when dealing with the motion blur of high-velocity shots. Consequently, traditional detectors trained on static images often fail to perform reliably when applied to the dynamic video feeds necessary for game analysis.

This project proposes the development of a Snooker Intelligent Analyzer & Shot Recommendation System. The system utilizes enhanced YOLO architectures, specifically adapting Hyper-YOLO concepts, to achieve high-accuracy real-time detection of all 21 snooker balls, distinguishing between similar colors using specialized color space transformations. Beyond simple detection, the project implements a Game State Analysis Engine (GSAE) that acts as a digital referee. By enforcing official World Professional Billiards and Snooker Association rules, the system automates score tracking and identifies legal shots, thereby removing the cognitive load of manual scoring and allowing players to focus entirely on their performance.

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1.1 Problem Statement

Snooker players, particularly beginners and intermediate enthusiasts, face challenges in shot selection and strategic decision-making during gameplay. Existing solutions in the sports technology market often fall short of addressing the specific needs of cue sports.

From a technical perspective, existing computer vision applications in sports have largely focused on "macroscopic" tracking, such as player movement in football or basketball. A comprehensive review by Abbas et al. (2022) indicates a distinct lack of detailed "object state recognition" required for strategic gaming applications. In snooker, the "state" is defined by complex, millimeter-level interactions between 22 rigid bodies. Existing mobile apps often function merely as digital scoreboards requiring manual input, which disrupts the player's "flow state" and fails to capture the rich spatial data such as cue ball position or shot difficulty, that is essential for genuine improvement.

The visual environment of a snooker table presents a "hostile" scenario for standard object detection algorithms. Snooker halls often feature variable lighting conditions that cause color distortion; for instance, under warm tungsten light, a Red ball can become visually indistinguishable from a Pink or Brown ball. Pan et al. (2024) highlight that distinguishing between similar-colored billiard balls requires advanced color space transformations and specialized neural network optimization to achieve the accuracy needed for reliable game analysis. Additionally, "ball occlusion" where balls cluster together poses a significant challenge for standard detectors, which often fail to separate individual instances within a pack.

The project also addresses the specific limitations of general-purpose detection models. Standard YOLO architectures, while fast, frequently struggle with "Small Object Detection" (SOD). A snooker ball on a full-size table occupies a minute fraction of the visual field, leading to feature dilution where the object's representation vanishes during the network's down-sampling process. Furthermore, the high velocity of snooker shots introduces motion blur. Hiemann et al. (2021) emphasize that traditional detectors trained on static images fail catastrophically when presented with blurred, fast-moving objects in video feeds. This necessitates the implementation of enhanced architectures, such as Hyper-YOLO, which utilize hypergraph computations to capture high-order correlations and preserve small-object features.

1.2 Project objectives

The system is designed to achieve the following key objectives:

1. To develop a real-time ball detection and classification system utilizing enhanced YOLO architectures (specifically adapting Hyper-YOLO concepts) to identify all 21 snooker balls with a detection accuracy of 95% or higher, while distinguishing between similar colors under variable lighting using specialized color space transformations.
2. To design a game state analysis framework that evaluates the current game situation in real-time, enforcing official World Professional Billiards and Snooker Association rules to automate score tracking, identify legal shots, and calculate remaining points on the table, thereby removing the cognitive load of manual scoring from the player.

1.3 Project Advantages & Contributions

1.3.1 Advantages

The primary advantage of this system is its ability to seamlessly automate the administrative aspects of the game, significantly enhancing the user experience. By utilizing a Game State Analysis Engine (GSAE) to automate score tracking and identify legal targets, the system removes the cognitive load associated with manual input, preventing disruptions to the player's "flow state". Furthermore, the system is designed for high accessibility and performance on mobile hardware; by converting the model to TensorFlow Lite (TFLite) and utilizing quantization, the application ensures low-latency execution on Android devices. This allows users to receive real-time visual feedback and augmented reality overlays directly through a standard smartphone camera, without the need for expensive, specialized broadcasting equipment.

1.3.2 Contributions

This project contributes to the field of sports technology by addressing the distinct lack of detailed "object state recognition" currently available for strategic gaming applications. Technically, it advances the application of computer vision in "hostile" visual environments by implementing enhanced architectures to solve specific issues related to "Small Object Detection" (SOD) and color distinction under variable lighting. By integrating these advanced detection capabilities with a rule-based logic engine, the project successfully democratizes professional-grade feedback, providing amateur players with a level of strategic analysis that

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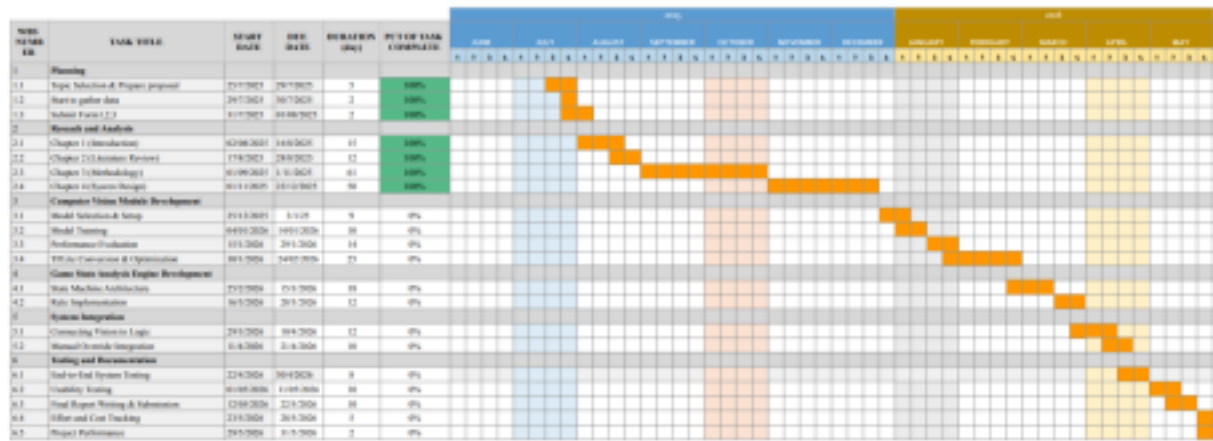
was previously inaccessible. Ultimately, the system serves as a bridge between complex computer vision concepts and practical, grassroots sports development.

1.4 Project Plan

ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
Phase 1: Literature Review & Environment Setup	Analysis of YOLO variants, billiards CV research, and video-based action recognition surveys, development environment configuration	

Phase 2: Enhanced YOLO Implementation	YOLO adaptation for snooker ball detection with optimization for small objects and real-time processing	
Phase 3: Color Classification System	Color classification achieving 99%+ accuracy for all 21 snooker balls using specialized neural networks	
Phase 4: Table Recognition & Geometric Analysis	Boundary detection and sub-centimeter coordinate mapping system implementation	
Phase 5: Ball Tracking Integration	Multi-frame ball tracking for occlusion handling and trajectory analysis	
Phase 6: Game State Analysis Framework	Complete snooker rules implementation with legal shot identification and scoring	
Phase 7: Real-time Processing Optimization	System optimization achieving >30 FPS processing with 95%+ accuracy maintenance	
Phase 8: Mobile Application Development	Implementation of computer vision system into Flutter mobile application with user interface design	
Phase 9: Validation & Performance Testing	Accuracy testing with real snooker scenarios and performance benchmarking	
Phase 10: System Integration & Documentation	Integration with teammate's ML module, technical documentation, and project demonstration	

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Gantt Chart

1.5 Project team and Organization

Student 1 Name (Backend): Ling Wan Sheng

Student 1 ID (Backend): 23WMR13217

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Student 2 ID (Frontend): 23WMR01649

Supervisor Name: Ms Nik Nur Adlin

Programme: RDS

1.6 Chapter Summary and Evaluation

This chapter introduced the fundamental premise of the project, identifying a clear gap in the sports technology market for amateur and intermediate snooker players. The problem statement effectively highlighted the technical limitations of existing computer vision solutions, specifically their inability to handle "hostile" environments characterized by variable lighting, ball occlusion, and small object detection (SOD) challenges. By proposing a Snooker Intelligent Analyzer that utilizes enhanced YOLO architectures and a Game State Analysis Engine (GSAE), the project establishes a robust technical direction. The objectives are evaluated as well-defined and ambitious, targeting high-accuracy detection (95%+) and the automation of complex rule enforcement to remove cognitive load from the player. This sets a strong foundation for the subsequent development of a system that democratizes professional-grade feedback.

Chapter 2

Literature Review

2 Literature Review

This chapter presents a comprehensive review of existing research and technologies related to computer vision applications in cue sports, with a particular focus on snooker and billiards analytics. It begins by examining the historical evolution of computer vision approaches, transitioning from traditional heuristic-based methods to modern deep learning paradigms. The chapter then reviews key challenges such as small object detection, ball occlusion, temporal tracking, illumination variance, and color classification under hostile lighting conditions. In addition, this chapter evaluates state-of-the-art object detection architectures, including successive YOLO variants and advanced frameworks such as Hyper-YOLO and slicing-based inference techniques, highlighting their relevance to snooker ball detection. Related work on dataset development, game state understanding, tracking consistency, and rule-aware analysis is also discussed. Collectively, this literature review establishes the theoretical foundation and technical justification for the design choices adopted in the proposed Snooker Intelligent Analyzer & Shot Recommendation System.

2.1 Evolution of Computer Vision in Cue Sports

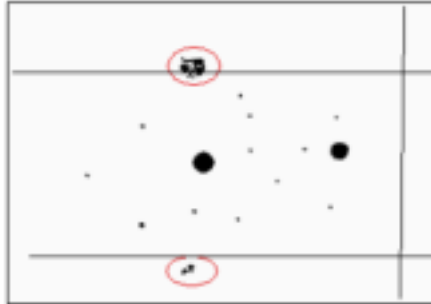
The application of computer vision to cue sports has historically been divided into two distinct eras: the heuristic era, characterized by rigid geometric assumptions, and the deep learning era, defined by data-driven feature representation.

2.1.1 The Legacy of Heuristic Approaches

Early attempts to digitize snooker, such as the Snooker Extraction and 3D Builder , relied heavily on classical image processing techniques. These systems typically employed Canny edge detectors followed by circle-based Hough transforms to identify balls. The fundamental assumption was that a snooker ball is a perfect circle of high contrast against a uniform green background. However, in real-world club environments, the "uniform" green cloth is often stained or unevenly lit, and balls particularly the green and blue ones can lack sufficient contrast for edge detection. Furthermore, specular reflections from overhead lighting often



(A)



(B)



(C)

, G., MacNamee, B.

2007)

(A) The original image. (B) The results of thresholding. (C) The final set of detected balls highlighted with green dots

While these methods were computationally inexpensive, their inability to handle semantic context (e.g., distinguishing a red ball from a red pocket liner) limited their utility to strictly controlled broadcast environments. They lacked the "cognitive" ability to understand the game state, treating each frame as an isolated geometric puzzle rather than a continuous event in a structured game.

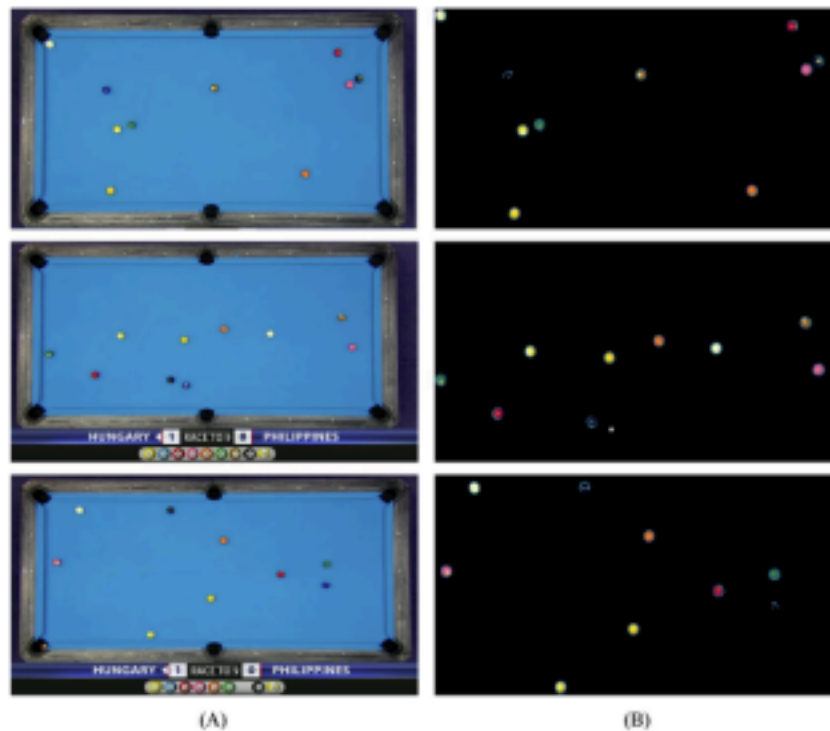
2.1.2 The Deep Learning Paradigm Shift

The inflection point in billiard analytics occurred around 2020, driven by the broad adoption of Convolutional Neural Networks (CNNs). A seminal study in robotic vision for 8-ball

billiards demonstrated that deep learning could overcome the specific limitations of heuristic methods, particularly in scenarios involving occlusion and motion blur. (Pan W et al, 2024) The researchers proposed a hybrid methodology that, while still utilizing some color space transformations for initial candidate proposals, relied on a modified Xception CNN for the critical task of pattern recognition. This network was optimized using an Improved Chaos African Vulture Optimization (ICAVO) algorithm, a meta-heuristic approach that fine-tuned the network's weights to escape local optima, thereby significantly enhancing classification

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accuracy between solid and striped balls. (H. Pan, 2023)



Final balls

segmentation This shift represented a fundamental change in how the problem was framed: ball detection

became a pattern recognition task rather than a geometric search. Deep networks could learn to identify a ball not just by its circularity, but by its texture, its specular highlight pattern, and its context relative to the table surface.

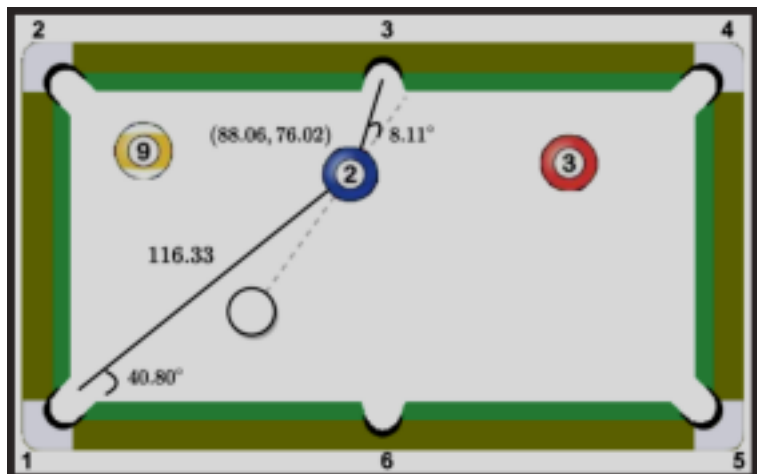
2.1.3 Dataset Scarcity and the Rise of "Game State" Understanding One of the most significant barriers to advancing snooker analytics has been the lack of publicly available, annotated datasets. Unlike the COCO or ImageNet datasets, which drove general object detection progress, specific datasets for cue sports were virtually non-existent before 2021. (Zhang, Q., et al 2024) Recent literature has begun to address this gap. A notable

contribution is the compilation of large-scale datasets from 9-ball tournaments, comprising over 58,000 annotated frames. (Zheng, A., & Yan, W. Q. 2025)

These datasets have enabled research into higher-order tasks beyond simple detection, such as:

Layout Prediction: Using architectures like the Billiards Layout CNN (BLCNN) to predict the outcome of a break shot or the probability of a clearance.

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Billiards Layout CNN

Layout Generation: Utilizing Generative Adversarial Networks (BLGAN) to create realistic training scenarios for players.

Similarity Search: The development of Deep Metric Learning models like BL2Vec, which allows systems to retrieve historical game situations that are strategically similar to the current table state.

This evolution is critical for the Snooker Intelligent Analyzer. It suggests that the "optical referee" need not be a passive observer but can actively interpret the game state, identifying distinct phases (e.g., the break-off, safety exchange, or clearance) to inform its processing logic.

2.1.4 Tracking and Temporal Consistency

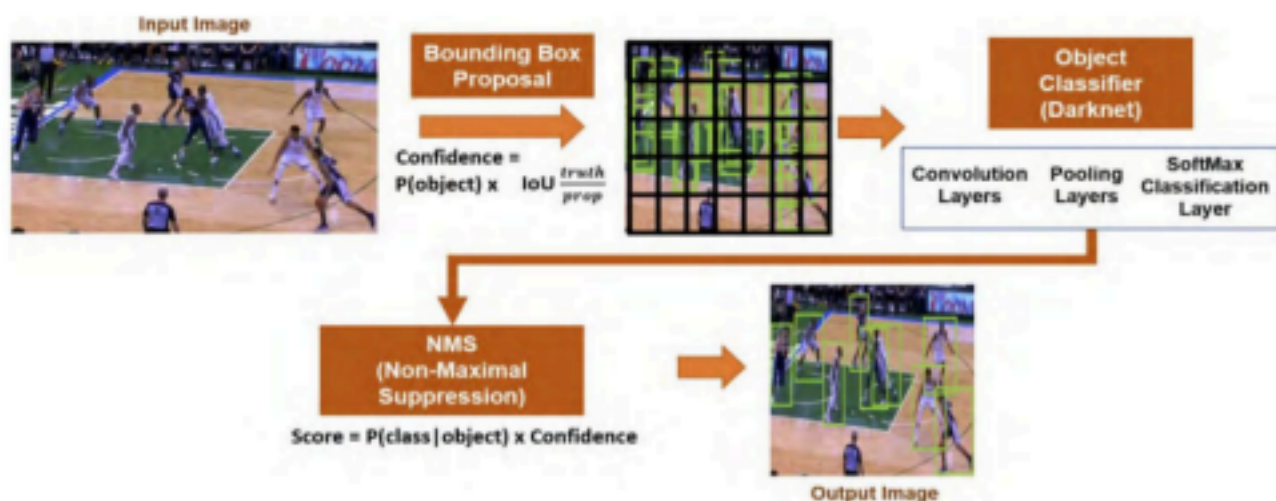
A persistent challenge in computer vision for sports is maintaining object identity over time, particularly during occlusion. In snooker, the "pack" of red balls presents a dense cluster where individual instances overlap, and players frequently block the camera's view while taking a shot.

The standard "tracking-by-detection" paradigm, often employing Kalman Filters or SORT (Simple Online and Realtime Tracking), has shown limitations when occlusions are prolonged. Recent research has proposed more robust solutions such as the Temporal Occlusion Tracking Network (TOTNet). TOTNet leverages 3D convolutions to process a sequence of frames simultaneously, using a visibility-weighted loss function to "remember" the location of balls even when they are fully obscured by a player. (Xu, et al, 2025)

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Tracking examples of TOTNet including scenarios with occlusion



Process of object detection and classification using YOLO

Similarly in basketball analytics, a domain with high transferability to snooker, researchers have successfully enhanced YOLO backbones with temporal context from adjacent frames, improving tracking accuracy under adversarial conditions by smoothing detection jitter.

(Mingsheng Xu, 2025) For a snooker analyzer, this implies that the system should not discard

detection history; rather, it should maintain a probabilistic model of the table state that persists even when visual data is temporarily interrupted.

2.2 Object Detection Architectures for Snooker

The core of any snooker vision system is the accurate detection of the table, balls, and pockets. This section details the specific architectures and techniques required, moving beyond generic object detection to domain-specific optimizations.

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2.2.1 YOLOv5

YOLOv5 served as the foundational standard for early real-time object detection systems due to its balance of speed and accuracy. Its architecture introduced the "Focus" layer (later evolved into 6x6 convolutions), which was critical for reducing CUDA memory usage during training while preserving essential image information. However, for a precision sport like snooker, YOLOv5 suffered from a significant limitation known as the "coupled-head" design. In this architecture, the neural network branches responsible for "localization" (finding where the ball is) and "classification" (determining the ball's color) shared the same parameters. In snooker, localization is geometrically simple because every ball is a circle, but classification is notoriously difficult due to lighting variations that make red and brown balls look identical. This parameter coupling meant that the network often struggled to optimize for color accuracy without compromising position, leading to frequent misclassification errors in "hostile" lighting environments. (Sapkota, R., & Karkee, M., 2025)

2.2.2 YOLOv8: The Anchor-Free Shift

YOLOv8 marked a revolutionary shift in computer vision by introducing a "Decoupled Head" architecture. By physically separating the regression (localization) and classification branches, the model could independently learn to detect the sharp edges of a ball without being confused by its texture or color. This separation significantly improved the system's ability to distinguish between "Red" and "Pink" balls, a common failure point in previous iterations. Additionally, YOLOv8 adopted an "Anchor-Free" detection strategy. Previous models relied on pre-defined anchor boxes, which required tedious manual tuning to account for perspective distortion (where spherical balls appear elliptical near the table edges). The anchor-free approach instead predicts the distance from the object's center to its boundaries, making it naturally robust to the wide-angle distortions common in table overhead cameras. Furthermore, the inclusion of the C2f (Cross-Stage Partial with 2 convolutions) module

improved gradient flow, allowing for deeper networks that could extract rich semantic features, such as distinguishing a white cue ball from a player's white shirt. (Sapkota, R., & Karkee, M. ,2025)

2.2.3 YOLO11: Efficiency and Feature Preservation

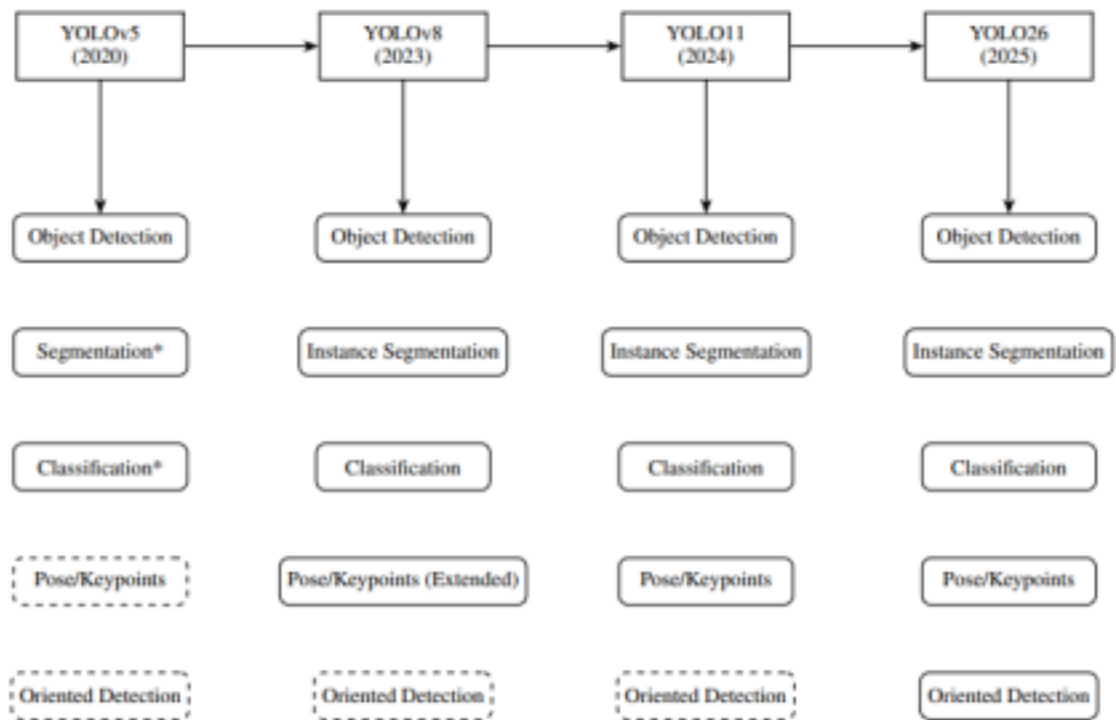
Released as an efficiency-focused update, YOLO11 refined the architecture specifically for detailed, small-object tasks. Its core innovation is the C3k2 block, which utilizes smaller kernel sizes to process visual data faster, optimizing parameter efficiency for mobile deployment. Crucially for your project, YOLO11 emphasizes the use of the P2 detection head. Standard detection heads often operate at low resolutions (strides of 8, 16, or 32), which causes small objects to "vanish" during down-sampling. The P2 layer operates at a much

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higher resolution (stride 4), preserving the fine-grained pixel data necessary to accurately detect snooker balls at the far end of the table. To further enhance accuracy, it incorporates the C2PSA (Cross-Stage Partial Spatial Attention) module. In a snooker frame, the majority of the image is empty green cloth; C2PSA allows the network to intelligently "suppress" this background noise and focus its computational resources solely on the relevant "action" areas—the clusters of balls and the player's stance. (Sapkota, R., & Karkee, M. ,2025)

2.2.4 YOLO26: The End-to-End Revolution

The YOLO26 architecture introduces an "End-to-End Revolution" that directly addresses the hardest problem in snooker computer vision: the "pack" of red balls. It achieves this through NMS-Free Inference via a one-to-one matching strategy. Traditional models use Non-Maximum Suppression (NMS) to delete duplicate boxes, but this often accidentally deletes valid red balls that are touching each other. YOLO26 eliminates NMS entirely, acting as a "silver bullet" for distinguishing individual balls within a dense cluster. It complements this with STAL (Small-Target-Aware Label Assignment), a training strategy that dynamically adjusts loss functions to prioritize small objects. This counteracts the natural bias neural networks have towards large objects (like the table or player), ensuring the model remains highly sensitive to the minute footprint of distal balls. Finally, the removal of Distribution Focal Loss (DFL) simplifies the output head, making the model lightweight enough to run efficiently on edge hardware like the NPUs found in commercial digital scoreboards. (Sapkota, R., & Karkee, M. 2025)



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Evolution of YOLO

Solid boxes = task supported natively, dashed boxes = task not supported.

2.3 Small Object Detection (SOD)

While deep learning has solved the general problem of object detection, recognizing snooker balls on a full-size table remains a specialized challenge categorized as Small Object Detection (SOD). On a standard 12ft x 6ft table, a 52.5mm snooker ball occupies a minute fraction of the visual field. In a 1080p video feed covering the full table, a single ball might be represented by a grid of fewer than 15 x 15 pixels.



Snooker Table and Balls

The detection of small objects presents unique computational difficulties due to insufficient feature representation and susceptibility to background interference, issues that are highly pertinent to the challenge of accurately tracking snooker balls. Recent advancements in

YOLO-based architectures for aerial and remote sensing scenarios offer robust solutions transferable to this domain. Standard detection models often struggle to balance performance and resource consumption for small targets, proposing "UAV-YOLOv8" to mitigate this by integrating a BiFormer attention mechanism and Wise-IoU (WIoU) loss. This approach enhances the model's focus on critical information while effectively suppressing background noise, achieving a 7.7% improvement in accuracy over baseline models. (Wang et al. 2023). Similarly, the issue of weak feature representation in remote sensing through "FFCA-YOLO," which utilizes Feature Enhancement Modules (FEM) and Spatial Context Aware Modules (SCAM) to enrich local context and establish global associations across channels. (Zhang et al. 2024)

These strategies, specifically the use of multi-scale feature fusion and lightweight attention mechanisms, provide a critical theoretical foundation for solving the SOD challenge in snooker, where targets are small relative to the frame and require precise localization against a dominant background.

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2.4 Advanced Architectures: Hypergraphs and Slicing The YOLO (You

Only Look Once) architecture represents a family of real-time object detection models known for their speed and efficiency. Beyond the standard YOLO progression, recent specialized research has introduced novel mechanisms to further boost small object detection performance. A development in 2024/2025 is Hyper-YOLO, which integrates hypergraph computations into the object detection pipeline. Standard CNNs rely on local convolution operations, which effectively model relationships between adjacent pixels (simple graphs). (Feng et al., 2024) However, a snooker game state is defined by "high-order" correlations; a cluster of red balls has a semantic relationship that transcends immediate pixel adjacency.

Hyper-YOLO introduces the Hypergraph-Based Cross-Level and Cross-Position Representation Network (HyperC2Net). In this framework, a "hyperedge" can connect multiple distant nodes in the feature map that share semantic similarities. This allows the network to perform "semantic collecting and scattering," aggregating information from all ball-like objects on the table to reinforce the detection confidence of each individual ball. Experimental results on the COCO dataset show that Hyper-YOLO-N outperforms YOLOv8-N by 12% in AP, a massive margin that highlights the efficacy of modeling these complex, non-local relationships. (Feng et al., 2024) For snooker, this could be the key to solving the occlusion problem in the "pack," where balls are adjacent and partially overlapping.

Another effective strategy for SOD is Slicing Aided Hyper Inference (SAHI). This technique involves partitioning a high-resolution input image into overlapping slices (e.g., dividing a 1080p frame into four 640 x 640 patches). Inference is run on each slice independently,

effectively zooming in on the balls and increasing their pixel resolution relative to the network's input size. The results are then stitched back together. (Rashad, 2025) While SAHI dramatically improves detection recall for small objects, it increases the computational load linearly with the number of slices. Recent studies suggest that using a lightweight model (like YOLOv8n) combined with SAHI can yield better accuracy than a heavy model (YOLOv8x) on a full image, offering a tunable trade-off for mobile applications. (Rashad, 2025).

2.5 Instance Segmentation

2.6.1 Video Instance Segmentation

Video Instance Segmentation (VIS) is a comprehensive computer vision task that integrates detection, segmentation, and tracking into a unified framework. The goal of VIS is to identify individual object instances in a video and maintain their unique identities across the entire sequence. (Yang et al. , 2019) Unlike standard image segmentation, which processes frames in isolation, VIS treats the video as a continuous volume. The model must recognize that an object

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(e.g., "Person A") appearing in the first frame is the same entity in the tenth frame, even if they have moved, changed pose, or been partially occluded. The final output is essentially a "space-time tube"—a sequence of binary masks assigned to a specific instance ID. Success is measured using the Space-Time Intersection-over-Union (IoU) metric, which evaluates both the pixel-level accuracy of the masks and the temporal consistency of the assigned IDs.

(a) Source image (b) Semantic segmentation (c) Instance Segmentation

Technically, VIS systems function through one of two primary architectural paradigms: online or offline. Online methods, such as MaskTrack R-CNN, process the video frame-by-frame, which is essential for real-time applications like autonomous driving. These systems typically extend standard image architecture (like Mask R-CNN) by adding a tracking branch; this branch computes a feature embedding for each detected instance, which is then compared against a memory bank of previous instances to link them based on visual similarity. Recent advancements like IDOL improve this by using contrastive learning to make these embeddings more discriminative, ensuring robust tracking even when objects look similar.

In contrast, offline methods process an entire video clip simultaneously, allowing the model to leverage future context to resolve ambiguities. Transformer-based models like VisTR and SeqFormer exemplify this approach by viewing VIS as a direct sequence prediction problem. Instead of matching frames sequentially, these models employ learned "instance queries" that attend to the entire space-time volume of the clip. This allows the network to aggregate information from multiple frames to generate a single, coherent representation for each object, effectively performing detection and tracking in a single end-to-end step without the need for complex post-processing or external tracking heuristics.

2.6 Classification Methodologies

2.7.1 Classification in Image and Video Processing

Classification serves as the foundational task in computer vision, designed to bridge the "semantic gap" between raw visual data and human understanding. In image processing, classification is the computational process of analyzing a static array of pixel intensities to assign a predefined label, such as "landscape" or "portrait," effectively teaching a machine to

recognize spatial patterns. Video classification significantly complicates this by introducing the temporal dimension; the system must not only recognize objects within individual frames but also interpret their movements and interactions over time to identify dynamic actions like "clapping" or "running". The justification for developing increasingly complex classification

algorithms lies in the inherent variability of visual data: a robust system must correctly categorize an object despite changes in lighting, viewpoint, occlusion, or background clutter, a feat that simple rule-based systems cannot achieve.

The mechanism of classification has shifted from rigid, manual engineering to flexible, data-driven learning. Historically, practitioners used "shallow learning" approaches where feature extraction was distinct from classification. Algorithms like the Histogram of Oriented Gradients (HOG) were manually designed to capture specific visual cues like edges or textures, which were then fed into classifiers like Support Vector Machines (SVM). However, this approach was limited by human ingenuity and domain specificity. Deep learning, particularly through Convolutional Neural Networks (CNNs), revolutionized this by unifying feature extraction and classification into a single "end-to-end" pipeline. CNNs work by passing images through layers of learnable filters; early layers detect simple geometric primitives (lines, curves), while deeper layers combine these to detect complex semantic objects (eyes, faces). This hierarchical learning structure is the primary justification for the superior accuracy of models like ResNet and EfficientNet over traditional methods, as they can automatically discover the most relevant features for any given task without manual intervention. (Haoran Cheng, 2025)

In the domain of video processing, classification mechanisms must explicitly model the passage of time to distinguish between actions that look similar in static frames (e.g., "sitting down" vs. "standing up"). To achieve this, researchers developed 3D Convolutional Neural Networks (such as C3D and I3D), which extend the 2D filters of standard CNNs into 3D "cubes". This allows the network to perform mathematical operations across the temporal axis, capturing motion and appearance simultaneously. Another prominent justification for video architecture design is the "Two-Stream" hypothesis, which mimics the human visual cortex by processing static appearance (what the object is) and motion dynamics (how the object moves, represented by optical flow) in two separate neural networks, later fusing their outputs for a more robust prediction. Most recently, Transformers (e.g., ViT, ViViT) have been introduced to address the locality limitations of CNNs. By treating images and videos as sequences of "patches" rather than grids of pixels, Transformers use self-attention mechanisms to capture global context, allowing the model to understand long-range dependencies across the entire image or video sequence. (Geeks, 2025)

2.7 Oriented Object Detection and Tracking

2.8.1 Oriented Object Detection in computer vision

Oriented Object Detection (OOD) is a specialized computer vision task designed to precisely

localize and classify objects that appear at arbitrary orientations, a requirement often found in aerial imagery (remote sensing), scene text detection, and industrial inspection. Unlike traditional horizontal bounding boxes (HBBs) which align strictly with image axes, OOD utilizes Oriented Bounding Boxes (OBBs) that include an additional angular parameter, typically represented as (x, y, w, h, θ) (Wang et al., 2025). This rotational capability is essential because using HBBs for diagonally oriented objects—such as ships in a harbor or vehicles in a parking lot—often captures excessive background noise and leads to significant overlap between adjacent objects, causing valid detections to be erroneously suppressed during post-processing (Xie et al., 2021).

The mechanism behind OOD works by integrating rotation-aware modules into deep learning architectures. Early innovations, such as the RoI Transformer, employed a learner to warp horizontal region proposals into rotated ones, ensuring that the features extracted for classification were spatially aligned with the object's actual orientation (Ding et al., 2019). More recently, the field has addressed the mathematical challenges of angular regression, specifically the "boundary discontinuity" problem where angular periodicity causes loss spikes by modeling bounding boxes as 2-D Gaussian distributions. Loss functions based on the Gaussian Wasserstein Distance (GWD) allow the network to minimize the probabilistic divergence between the predicted and ground truth distributions, providing a smooth and differentiable optimization landscape even for square-like objects where orientation is ambiguous (Yang et al., 2021).

Oriented Object Tracking (OOT) extends these detection capabilities into the temporal domain to monitor object trajectories across video sequences. OOT systems typically operate on a "tracking-by-detection" paradigm, where an OOD model localizes objects in each frame, and a tracker associates them with existing trajectories using motion models like the Kalman filter, which must be adapted to handle angular velocity and cyclic orientation data. To maintain consistent identities in dynamic environments, these trackers utilize specialized association metrics, such as SkewIoU or Gaussian-based distances, which are robust to the complex motions and ego-motion (camera movement) often present in drone or autonomous driving footage .

2.8 Illumination Invariance and Color Classification

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In snooker, object detection is only half the battle; the system must also classify the balls into 8 distinct categories (Cue, Red, Yellow, Green, Brown, Blue, Pink, Black). This is fundamentally

a color classification problem. However, the visual environment of a snooker hall is complex, characterized by variable lighting conditions, uneven illumination, and colored shadows cast by the balls themselves.

A study on sports ball visibility under different CCTs (Correlated Color Temperature) revealed that color perception is frequency-dependent. White balls (high reflectance across the spectrum) exhibit better contrast under lower CCT (warmer, 3000K) lighting due to the relative darkness of the background, whereas yellow balls are more distinct under higher CCT (cooler, 5000K+) lighting. This implies that a static classifier will inevitably fail as the lighting environment changes. The distinction between a Red and a Pink ball, or a Brown and a Red ball, is particularly susceptible to the "metamerism" effect where different spectral power distributions result in identical perceived colors. (Andre Bakker, 2025)

To address this, modern systems are moving toward Deep Learned Color Constancy. Rather than relying on fixed thresholds in HSV space, recent approaches utilize foundation models to learn "bias-invariant" representations. (Sarridis et al., 2024) In the context of snooker, the system learns to associate the relative color shifts of all balls on the table. For example, if the White ball appears yellowish due to lighting, the system infers that the "Pink" ball will also be shifted toward orange and adjusts its classification boundary dynamically. This mimics the human visual system's opponent-color mechanism, which has been shown to be far more robust than raw pixel analysis. (Flachot A et al., 2022)

2.9 Chapter Summary and Evaluation

This chapter critically reviewed existing literature relevant to the application of computer vision and artificial intelligence in snooker and cue sports analytics. The evaluation of early heuristic-based approaches revealed that while they were computationally efficient, they lacked robustness in real-world environments due to sensitivity to lighting variation, occlusion, and limited semantic understanding. The transition to deep learning-based methods marked a significant advancement, enabling more reliable ball detection, classification, and contextual interpretation of game states.

The analysis of modern object detection architectures, particularly the evolution of YOLO models, demonstrated that recent innovations such as decoupled heads, anchor-free detection, high-resolution feature preservation, and small-object-aware training strategies are highly suitable for snooker scenarios. Advanced techniques including Hyper-YOLO, SAHI, and attention-based feature enhancement were identified as effective solutions to persistent challenges such as clustered red balls and distant small targets.

Furthermore, the review of tracking methodologies, video instance segmentation, and temporal consistency models highlighted the importance of maintaining object identity across frames in dynamic and occluded conditions. Studies on illumination invariance and color constancy emphasized that accurate ball classification cannot rely solely on static color thresholds, reinforcing the need for adaptive and learning-based approaches.

Overall, this chapter establishes a strong theoretical and technical foundation for the proposed system. The evaluated literature validates the feasibility of integrating enhanced YOLO-based detection, robust tracking, and rule-aware game state analysis. These insights directly inform the architectural and methodological decisions presented in subsequent chapters, ensuring that the proposed solution addresses both practical snooker-specific challenges and current research gaps in sports computer vision.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis This chapter

outlines the systematic approach adopted for the development of the Snooker Intelligent Analyzer, detailing the software development lifecycle, technical tools, and specific system requirements. The project utilizes a hybrid Agile methodology, incorporating Extreme Programming (XP) principles to accommodate the iterative nature of training deep learning models and the need for rapid feedback loops in computer vision development. The chapter further defines the project's technical architecture, justifying the selection of Python and Optimized YOLO for the backend detection modules, and Flutter for the high-performance mobile frontend. Finally, it establishes the scope of the system through a comprehensive breakdown of functional and non-functional requirements, ensuring that the final deliverable meets the performance benchmarks of real-time sports analytics, such as sub-100ms latency and high detection accuracy.

3.1 Development Method Used

3.1.1 The Agile Paradigm in Computational Vision (Extreme Programming) The

technical foundation of this project is built upon a hybrid Agile methodology, specifically tailored for artificial intelligence and computer vision (CV) initiatives. Traditional linear models, such as the Waterfall approach, are increasingly viewed as inadequate for deep

learning projects due to the exploratory nature of model training and the inherent unpredictability of data-driven results. In computer vision, requirements often evolve as the model encounters edge cases in varied lighting or perspective distortion, necessitating a framework that allows for rapid pivoting and continuous feedback loops.

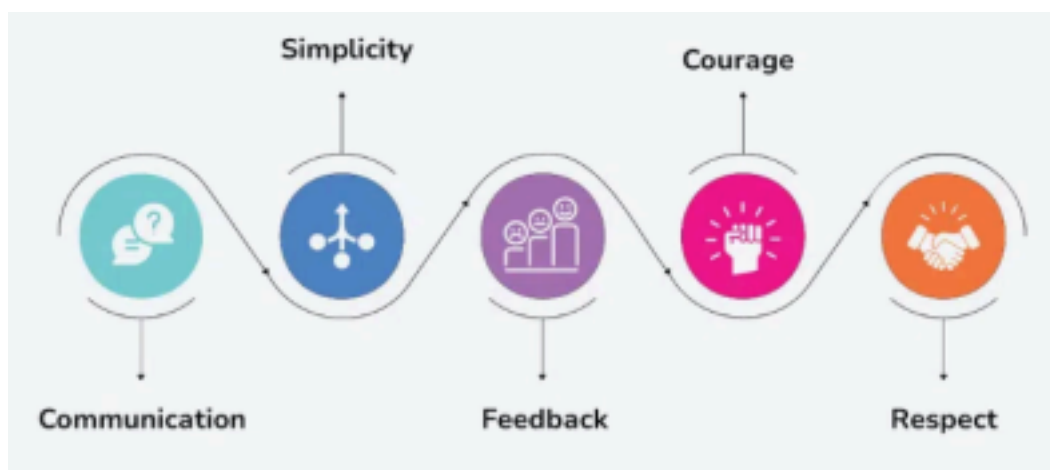
The adoption of Agile methodologies incorporating elements from Scrum and Extreme Programming (XP) facilitates an environment of faster experimentation and high adaptability. The development team can address specific technical hurdles, such as optimizing "Small Object Detection" (SOD) or refining the homography matrix, in isolated, measurable increments. This iterative approach ensures that model evaluation metrics like Mean Average Precision (mAP) and Intersection over Union (IoU) are assessed early and often, preventing the "black box" failure common in long-term AI research projects.

Because the system relies on complex mathematical transformations and high-frequency data streams, the technical team adopts XP values—Communication, Simplicity, Feedback, Courage, and Respect—to maintain a high standard of software engineering. Simplicity is particularly emphasized to avoid "over-engineering" the initial CV module, focusing first on the most efficient YOLO architectures (like YOLOv8n) before considering more computationally expensive hypergraph computations if accuracy targets are not met. The

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immediate feedback provided by unit testing and automated validation scripts ensures that changes to the model's quantization or the homography calculation do not introduce regressions in detection fidelity. (Jesse Anglen, 2025)



Values of Extreme Programming (XP)

3.2 Project Development Phases (Extreme Programming XP)

Phase 1: Planning

The project begins with the planning phase, where the primary objective is to define user needs

through concise "User Stories". In this phase, clients (amateur and intermediate snooker players) articulate requirements such as the need for affordable coaching alternatives, real-time ball detection, and automated scoring to preserve their "flow state". The technical team then calculates the effort required for each story such as the complexity of training a YOLO model for small object detection and schedules releases according to business priority and technical feasibility.

Phase 2: Design

The design phase focuses on creating only the essential architecture needed to fulfill the current user stories, adhering to the principle of simplicity to avoid over-engineering the initial Computer Vision Module. A common analogy or story the "Optical Referee" is utilized to help all stakeholders understand the overall system architecture, which consists of the detection backbone, geometric mapping engine, and the rule-enforcement logic. This keeping the design straightforward and clear ensures that the system remains performant on resource-constrained Android devices.

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Phase 3: Coding

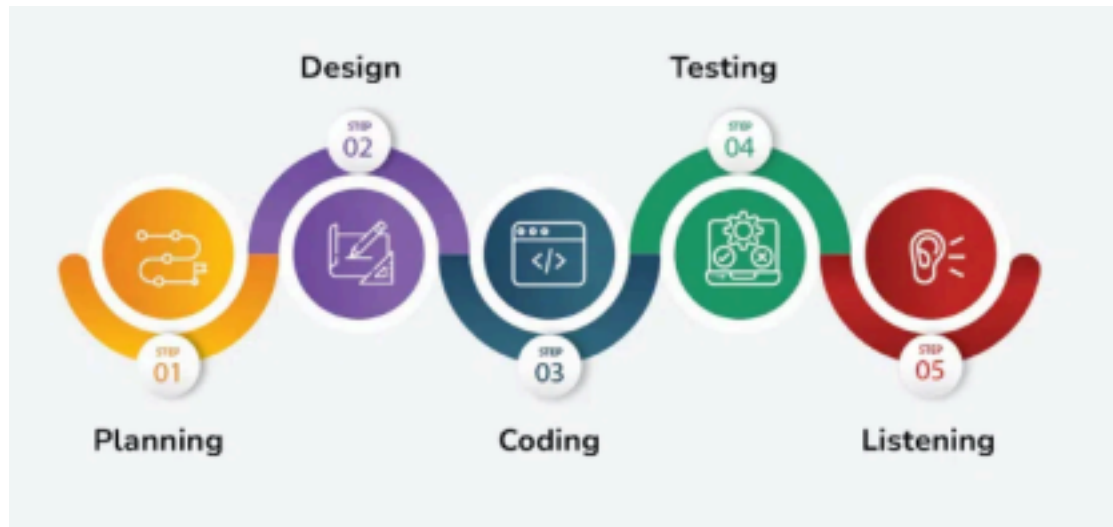
XP promotes pair programming, where two developers work together at one workstation, which is particularly advantageous for implementing complex snooker-specific algorithms like object detection and shot difficulty calculation. This collaborative environment enhances code quality and facilitates knowledge sharing across the computer vision and Flutter domains. The team employs Test-Driven Development (TDD) by writing tests before coding the rule-based Game State Analysis Engine (GSAE) to ensure functionality from the start. Code is frequently integrated into a shared Git repository with automated tests to catch integration issues early.

Phase 4: Testing

Testing is a cornerstone of the XP lifecycle, emphasizing both automated unit tests and customer-facing acceptance tests. Unit tests are automated to check specific features like the accuracy of RGB-to-HSV transformations and the sub-centimeter precision of coordinate mapping. Acceptance tests are conducted by snooker players to ensure that the overall system from the real-time camera overlay to the final score calculation meets the initial requirements and provides a seamless user experience.

Phase 5: Listening

In the listening phase, the development team seeks regular feedback from customers (the snooker community) to ensure the product continues to meet their needs and to adapt to any environmental changes. Feedback regarding the system's performance under different lighting conditions or on varied table cloth colors allows the team to refine data augmentation strategies and improve the model's "illumination invariance" in subsequent iterations. This continuous loop ensures the analyzer remains a robust tool for grassroots skill development.



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Life Cycle of Extreme Programming (XP)

3.3 Technical Stack and Tool Justification

Category	Tool / Technology	Description and Purpose
Back end	Python	Python serves as the system's primary backend environment, chosen for its status as the industry standard for AI and data science. Python is used to train and refine the YOLO detection backbone, leveraging libraries like PyTorch and OpenCV for advanced image preprocessing and color space manipulation. The core Game State Analysis Engine (GSAE) is initially developed and tested in Python before its rules-based logic is integrated into the final application environment.

Front end	Flutter	Flutter is utilized for the frontend to provide a "cross-platform power" and "Native-like performance". Fast Iteration: The "Hot Reload" feature speeds up development by 40-60%, which is vital for a student team with tight deadlines. High-Performance Rendering: Utilizing the Skia graphics engine allows for smooth, low-latency UI updates on top of the live camera stream, ensuring no "UI jank" occurs during real-time tracking.
Model	Optimized YOLO	The detection module uses an Optimized YOLO architecture (specifically YOLOv8n or YOLO11n) to achieve the state-of-the-art balance between inference speed and accuracy on mobile hardware.
	TFLite	Mobile Optimization: The model is converted to TensorFlow Lite (TFLite) to enable efficient execution on Android devices. Quantization: To meet memory and power constraints, the model undergoes INT8 or Float16 quantization, which significantly reduces the model's footprint while maintaining millimeter-level localization accuracy.

3.4 System Modules and Architectural Design

The Snooker Intelligent Analyzer is structured into three primary modules, each handling a distinct stage of the data pipeline.

Module 1: Computer Vision and Object Detection

This module is the "eyes" of the system, responsible for converting raw camera frames into semantic objects. It utilizes the optimized YOLO backbone to extract multi-scale features, allowing it to detect small balls at the far end of the table as effectively as those in the foreground. To handle "ball occlusion" in dense clusters (the "pack" of reds), the module implements advanced NMS (Non-Maximum Suppression) logic to distinguish between overlapping instances without generating false negatives.

Module 2: Game State Analysis Engine (GSAE)

The GSAE serves as the "brain" that interprets spatial data through the framework of official snooker rules. It receives the standardized metric coordinates from Module 1 and processes them through a state machine that tracks the match progression.

Rule Enforcement: The engine enforces official WPBSA rules to identify the "ball on" (the legal target), validates if the first ball hit was legal, and detects foul conditions such as potting the cue ball.

Automated Scoring: By analyzing the removal of balls from the table and identifying their colors, the GSAE automates score tracking, removing the cognitive load of manual input from the player.

Strategic Feedback: It calculates the remaining points on the table and identifies the current break, providing the baseline data for future shot recommendations.

Module 3: App Implementation

The app implementation utilizes the Flutter framework to provide cross-platform compatibility and high-performance rendering through its native Skia engine. Real-time visual data is captured using the Flutter camera plugin's startImageStream function, while computationally intensive tasks like image preprocessing and model inference are offloaded to background Dart isolates to prevent UI lag and maintain a high frame rate. Detected balls and table boundaries are then visualized for the user by rendering dynamic bounding boxes and labels directly over the live camera preview using a Stack widget.

3.5 Requirements Analysis

A successful sports analytics system must adhere to a strict set of functional and non-functional requirements. These requirements define the boundaries of the system and ensure that the final product fulfills the strategic objective of democratizing professional-grade feedback.

3.5.1 Functional Requirements Specification

Functional requirements describe "what" the system does—the specific features and behaviors that allow users to achieve their goals.

ID	Feature Name	Requirement Description
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FR1	Real-time Ball Detection	Identify and track all 22 balls (15 reds, 6 colors, and 1 cue ball) with a target accuracy
FR2	Table Pocket Detection	Automatically recognize the six pocket locations to define the playing area boundaries
FR3	Ball Classification	Categorize detected balls into eight distinct color classes using specialized HSV color space transformations
FR4	Score Tracking	Enforce official WPBSA rules to autonomously update the player's break and total score
FR5	App implementation	A Flutter-based mobile application that performs on-device TFLite inference

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3.5.2 Non-Functional Requirements Specification

Non-functional requirements specify "how" the system operates, addressing quality attributes like performance, security, and scalability.

ID	Feature Name	Requirement Description
NFR1	Accuracy	Object localization must maintain an IoU threshold of at least 0.5 to be considered a true positive.
NFR2	Portability	The application must function on Android 10+ and support diverse screen sizes (4.7" to 6.7").
NFR3	Security	All processing must be done on-device to ensure user privacy and data protection.
NFR4	Performance	The system must process camera frames and return results in < 100ms

3.6 Chapter Summary and Evaluation

This chapter outlined the methodological framework and technical specifications governing the project's development. The adoption of the Agile Extreme Programming (XP) methodology is evaluated as highly suitable for this initiative; unlike traditional linear models, XP's emphasis on feedback and iterative cycles directly addresses the exploratory nature of

training deep learning models where data characteristics often dictate development direction. The selection of the technical stack is robust, leveraging Python for its dominance in AI training and Flutter for its cross-platform performance capabilities. Furthermore, the decomposition of requirements into Functional (e.g., Real-time Ball Detection) and Non-Functional (e.g., <100ms processing time) provides a clear set of benchmarks. This structured approach ensures that the "Mobile Optimization" constraints, such as TFLite quantization, are prioritized early to meet the hardware limitations of standard smartphones.

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Chapter 4

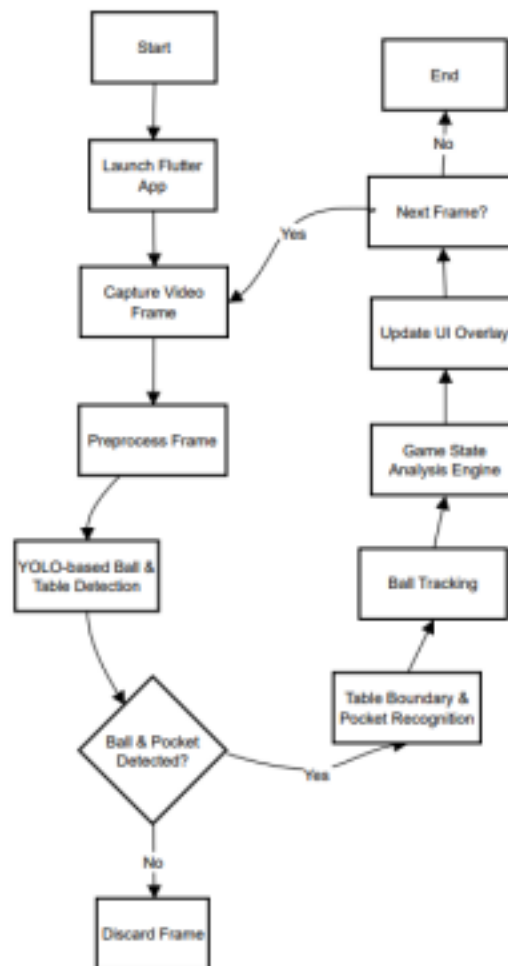
System Design

4 System Design

This chapter details the architectural blueprints and logical design of the Snooker Intelligent Analyzer, translating the requirements defined in Chapter 3 into executable system specifications. It begins by mapping the System Workflow, illustrating the continuous processing loop from video capture to the Augmented Reality (AR) overlay. The chapter then presents the User Interface (UI) Design, focusing on a single-view dashboard that prioritizes the live camera feed while providing a "Smart Scoreboard" for real-time feedback. A critical component detailed herein is the Game State Analysis Engine (GSAE), which functions as a "Digital Referee" using a state machine architecture to manage complex snooker rules, such as the transition between "Red Ball Active" and "Colour Nomination" phases. Finally, the chapter visualizes user interactions through Use Case diagrams, defining how players initiate analysis, upload footage, or trigger manual overrides.

4.1 Process Design

4.2 system workflow



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Process Design Diagram (Flowchart)

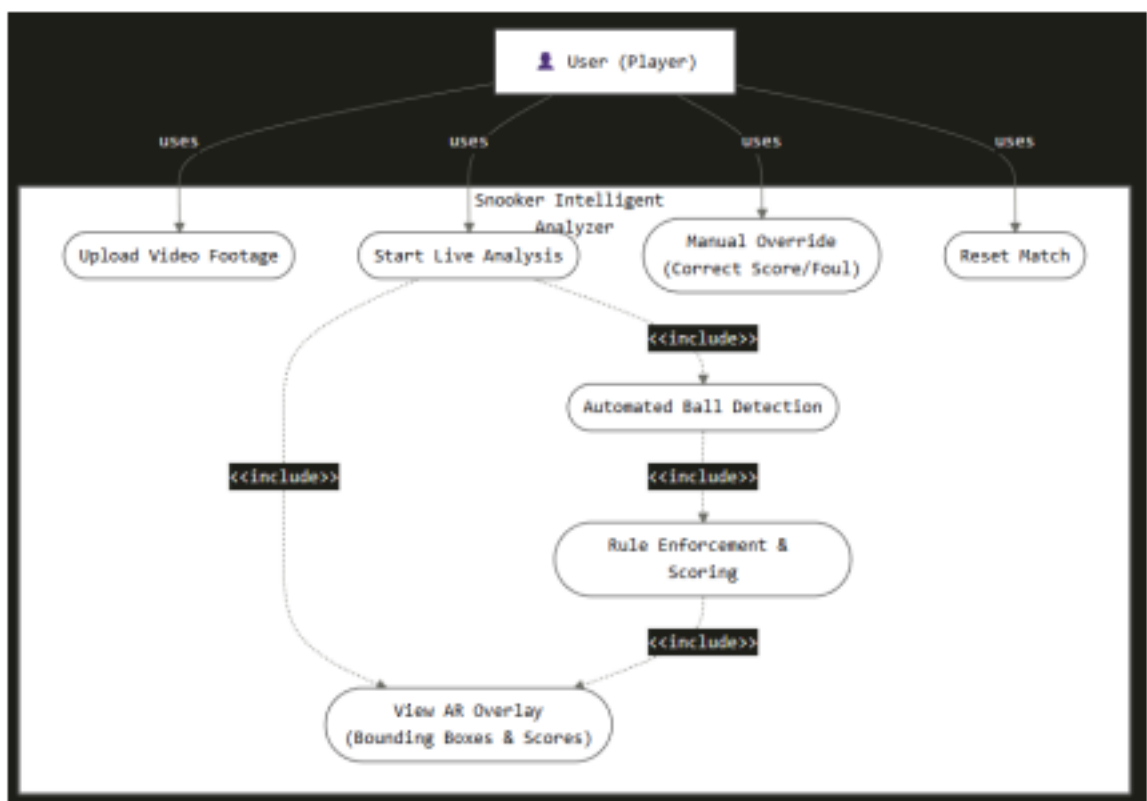
The system workflow initiates with the Launch of the Flutter Application, which initializes the camera sensors and loads the necessary neural network models into memory. Once the user starts a session, the system enters a continuous loop where it begins to Capture Video Frames from the device's camera stream. Each raw image is immediately passed to the Preprocess Frame stage, where it is resized and normalized to meet the input requirements of the deep learning model.

The core analysis begins with the YOLO-based Ball & Table Detection module. This step attempts to identify all relevant objects, specifically snooker balls and the table structure within the current frame. A critical validation step follows: Ball & Pocket Detected?. If the system fails to identify these essential elements with sufficient confidence (e.g., due to motion blur or camera obstruction), the process moves to Discard Frame, effectively skipping the current image to conserve processing power and waiting for the next clear video frame.

If valid objects are detected, the data flows into Table Boundary & Pocket Recognition, where the system calculates the table's geometry to establish a reference coordinate system. This spatial data is then fed into Ball Tracking, which correlates the current positions of the balls

with their positions from previous frames to determine velocity and trajectory. This trajectory data is consumed by the Game State Analysis Engine, which applies the rules of snooker to determine if a pot occurred, if a foul was committed, or if the turn has ended. Finally, the system executes an Update UI Overlay to display real-time AR bounding boxes and scores to the user. The cycle concludes with a check for Next Frame?; if the session is active, it loops back to capture the next image, otherwise, the process terminates at End.

4.3



Use Case

Use Case Diagram

Upload Video Footage This use case represents the system's post-game analysis capability. It allows the user to select and import pre-recorded video files (such as MP4 or MOV formats) from their device storage into the application. Once uploaded, the system processes the footage frame-by-frame using the same computer vision logic used in live tracking. This function is essential for players who wish to review their technique or analyze matches that were recorded previously, decoupling the analysis process from real-time gameplay.

Start Live Analysis This is the primary function of the application, where the user initiates a real-time tracking session using the device's camera. As indicated by the <<include>> relationships in the diagram, triggering this use case automatically executes a chain of dependent internal processes. First, it invokes Automated Ball Detection to locate the snooker balls and table boundaries. The spatial data from this detection is then passed to Rule Enforcement & Scoring, which applies the specific logic of the "Game State Analysis Engine" to valid shots. Finally, the processed data culminates in the View AR Overlay function, which superimposes bounding boxes and live scores directly onto the user's camera feed.

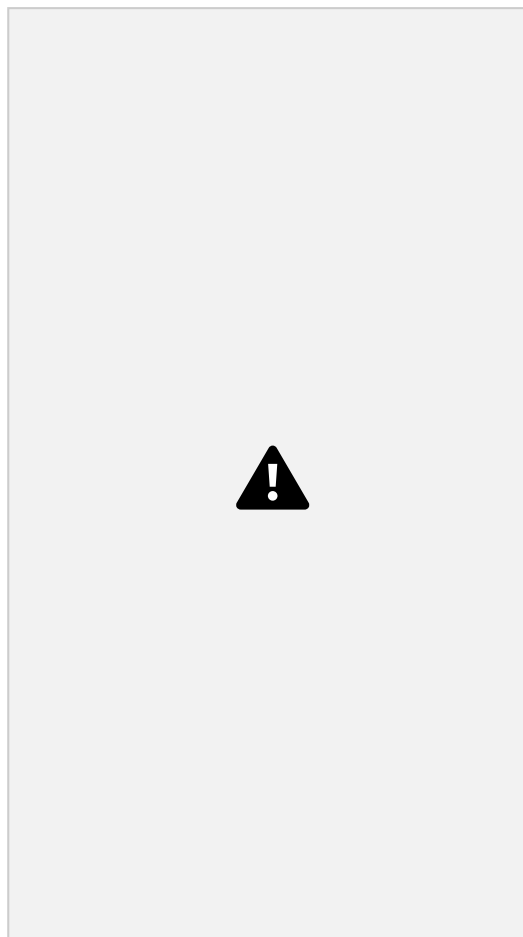
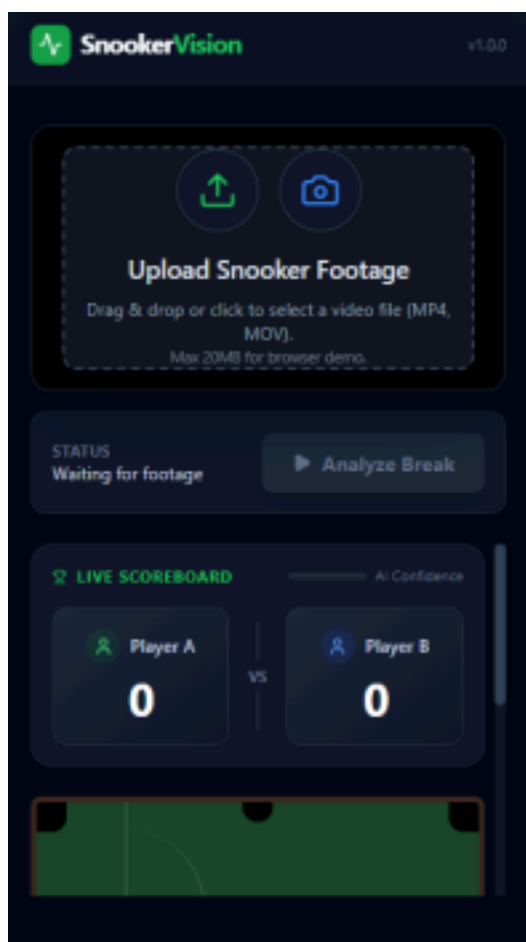
Manual Override (Correct Score/Foul) This use case provides a necessary human intervention mechanism to ensure data accuracy. Snooker environments can present "hostile" conditions

such as variable lighting or ball clusters that may occasionally confuse the computer vision algorithms. The Manual Override allows the user to temporarily pause the autonomous tracking to correct the score or manually input a complex foul (e.g., a "Miss" or "Free Ball" scenario). This ensures that the digital scoreboard remains accurate even if the AI misinterprets a specific game event, keeping the user in control of the final match data.

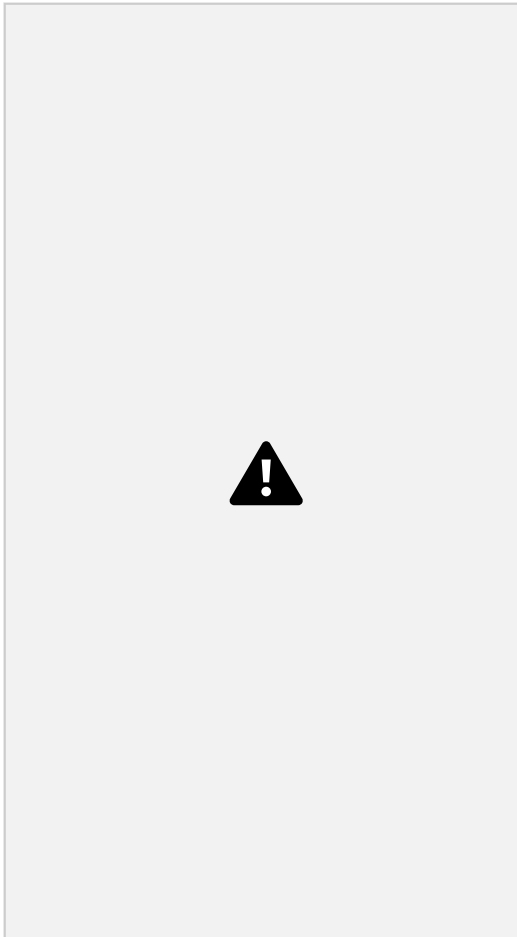
Reset Match The Reset Match use case serves as a system utility that allows the user to clear the current game session without exiting the application. When triggered, this function flushes all temporary variables stored in the Game State Analysis Engine, including the current break, total scores for both players, and the foul history. This returns the system to the initial "AwaitingBreak" state, allowing the players to immediately begin a new frame with a clean slate.

4.4 UI design

UI Layout and Component Description



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The user interface of the Snooker Intelligent Analyzer is designed as a vertically stacked, single-view dashboard that prioritizes the live video feed while providing comprehensive game data . The screen layout is divided into three logical zones: the Input/Navigation Header, the Augmented Reality Viewport, and the Score Display Section.

The Score Display Section (Center) is dedicated to the "Smart Scoreboard". This section is used to display the current frame scores that are automatically updated by the system. Clicking the scoreboard will open a control bar including buttons like foul, reset match, and manual mode allowing the user to override the autonomous tracking during complex rule scenarios, ensuring the digital scoreboard always maintains accurate score and result.

The Augmented Reality Viewport (Bottom) Occupying the majority of the screen space is the live camera feed, which serves as the primary canvas for the Computer Vision module. This area functions as an Augmented Reality (AR) window where the system visualizes its internal logic.

4.5 Game State Analysis Engine Design

The Game State Analysis Engine functions as the "Digital Referee." The primary design challenge here is managing the complexity of Snooker rules, which involve state-dependent legality (e.g., potting a Red is only legal in specific phases).

4.3.1 Game State Machine Logic

1) State: AwaitingBreak

Entry: Game initialization or Re-rack.

Legal Target: Red Pack.

Transition: If Red potted -> State: ColourNomination. If Miss/Foul -> State: FoulHandler.

2) State: RedBallActive (The "Red" Phase)

Constraint: Any Red ball is a legal target.

Transition (Pot Red): Score +1. Transition to State: ColourNomination.

Transition (Foul): Calculate penalty (min 4). Transition to State: FoulHandler.

3) State: ColourNomination (The "color ball choice" Phase)

Description: The player has potted a red ball and must now pot any colour ball.

Input: Yellow Ball = +2 Points

Green Ball = +3 Points

Brown Ball = +4 Points

Blue Ball = +5 Points

Pink Ball = +6 Points

Black Ball = +7 points

Transition to State: RedBallActive

Transition (Foul): Calculate penalty (min 4). Transition to State: FoulHandler.

4) State: ClearancePhase

Description: All Reds gone. Colours must be potted in sequence: Yellow(2) -> Green(3) -> Brown(4) -> Blue(5) -> Pink(6) -> Black(7).

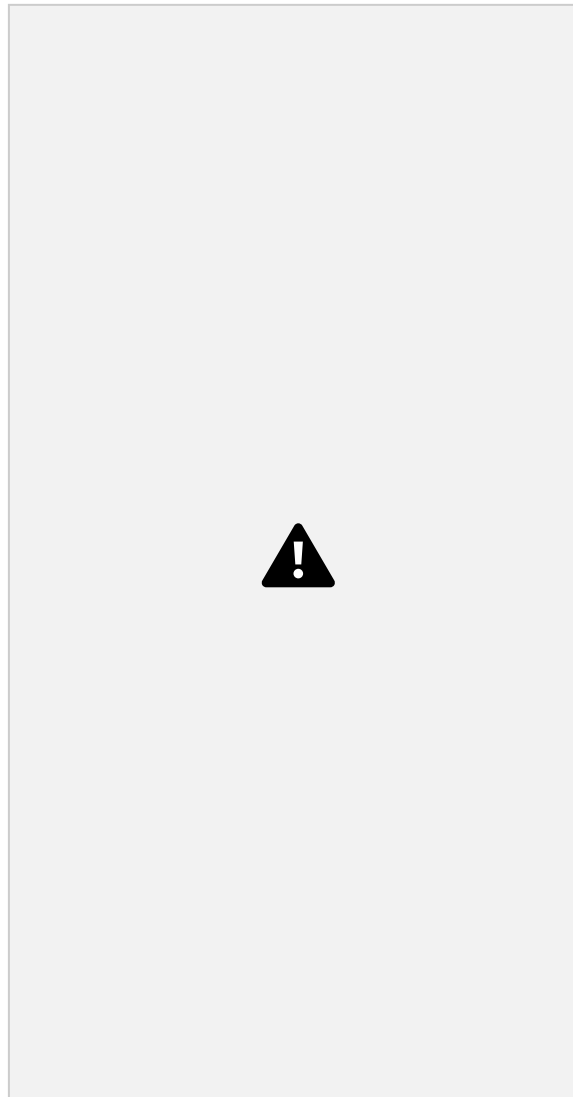
Logic: Dynamic target updating. Potting Yellow automatically sets legal targets to Green.

5) State: FoulHandler

Calculates penalty points based on the value of the target ball vs. the actual ball hit (max value rule).

Determines if a "Miss" should be called (Failing to hit any other ball, Potting the cue ball).

Checks for "Free Ball" scenario (if the opponent is snookered after the foul).



Snooker balls point

4.6 Chapter Summary and Evaluation

This chapter detailed the logical and architectural design of the Snooker Intelligent Analyzer, translating the requirements into executable system blueprints. The Process Design is evaluated as efficient for real-time applications; specifically, the inclusion of a validation step to "Discard Frame" when detection confidence is low effectively preserves processing power and prevents erratic behavior. The core logic mechanism, the Game State Analysis Engine (GSAE), successfully employs a state machine architecture (e.g., distinct states for "RedBallActive" vs. "ColourNomination"). This approach solves the complexity of "state-dependent legality" inherent in snooker rules, which simpler linear logic could not handle. Additionally, the UI Design balances automation with user control by integrating a Manual Override, which acts as a crucial fallback mechanism for "hostile" visual

environments, ensuring the system remains reliable even when the AI faces occlusion or lighting challenges.

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